TAGUIG CITY UNIVERSITY
COLLEGE OF INFORMATION AND COMPUTING TECHNOLOGY

A Risk-Based Two-Factor Student Verification and Event Attendance Management System

Using Near Field Communication (NFC), Personal Identification Number (PIN) Authentication, and Quick Response (QR) Credential Validation

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Pre-oral Evidence

Recommendation	Status
01-02 Checklist/visuals	Included
03 Raw responses	Packaging pending
04 Test evidence	Selected tests documented
05 UI appendices	Updates pending

Complete evidence package remains pending.

Pre-oral Comparison Requirements

Recommendation	Status
06-07 Manual baseline	Included
08 Respondent counts	Included
09-10 Same participants	Different groups used
16 Use case	Mode alignment pending
17 Survey t-test	Welch test included

Pre-oral System Requirements

Recommendation	Status
11 Login timestamp	Display needs verification
12 Student timestamps	Implemented
13 Result retention	10 seconds
14 Admin purge	Local cleanup only
15 Data destruction	Cloud retention unresolved

Problem and Objectives

Paper attendance separates checks from records.

Integrate credential checks with attendance records.

Configure Fast, Standard, and High-Security modes.

Handle entry, exit, and event eligibility.

Describe performance and perceived quality.

Compare manual and system survey ratings.

Configured Verification Modes

Mode	Credentials
Fast	NFC
Standard	NFC + PIN
High-Security	NFC + PIN + QR

Operators select the verification mode.

QR validates NFC-linked credential consistency.

Research Design and Respondents

Developmental, descriptive-evaluative research

ISO/IEC 25010-based survey questionnaires

System students	70
Manual students	70, separate group
Security personnel	5
Faculty/organizer/admin	5

Controlled Observation

Batch	Students	Rounds per mode
Aug 12	40	20 + 20
Aug 13	30	20 + 10

70 participants completed all three modes.

20 cards reused after reset exits.

Fixed order: Fast, Standard, High-Security.

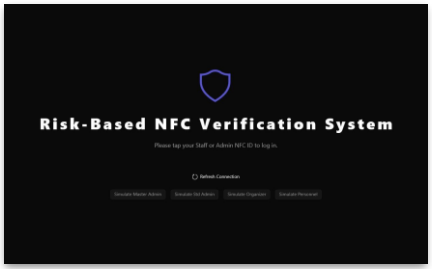
System Architecture

Component	Role
ESP32 + PN532	NFC UID reading
WinUI 3	Rules and timestamps
MySQL	Local record storage
Firestore	Cloud sync and backup

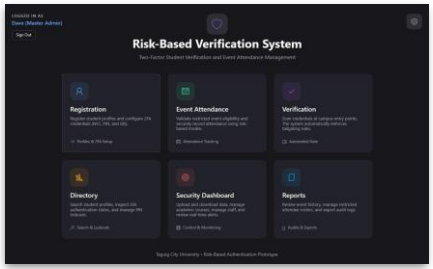
Later additions require separate evaluation.

Prototype Interfaces

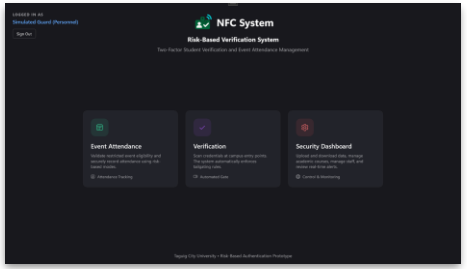
Authentication



Admin Dashboard



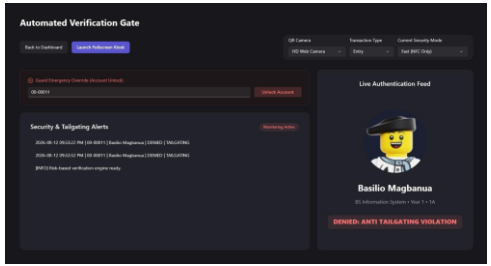
Security Dashboard



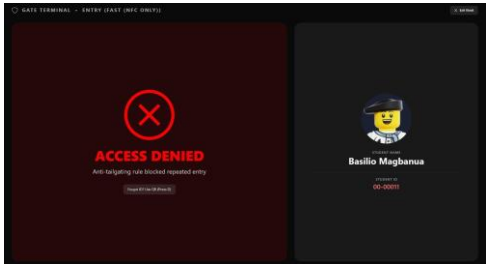
Event Organizer



Active Monitoring



Student Kiosk



Prototype screenshots. Current build in demonstration.

Statistical Treatment

Means and respondent-level sample SDs

Welch independent-samples t-test

Two-tailed threshold: alpha = 0.05

H0: Equal overall mean survey ratings.

H1: Different overall mean survey ratings.

t = mean difference / SE

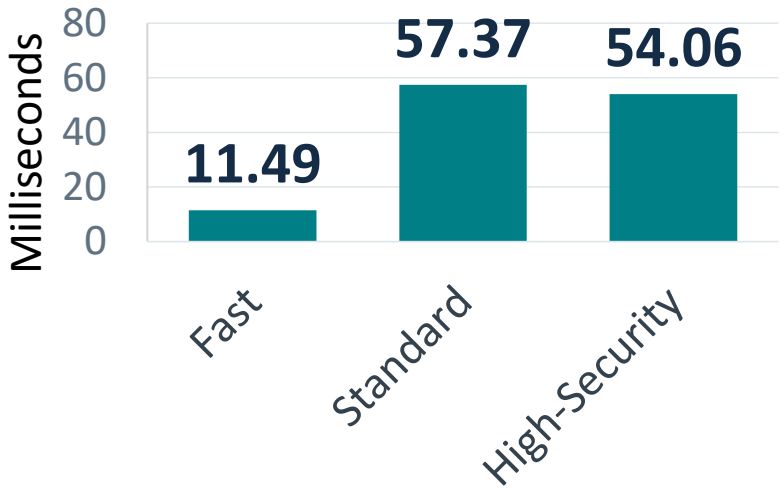
Records and Selected Test Cases

Mode	Entries	Resets	Denials	Total
Fast	70	70	2	142
Standard	70	70	12	152
High-Security	70	70	4	144
Total	**210**	**210**	**18**	**438**

16 listed PASS cases, limited scope.

Internal Latency (ms)

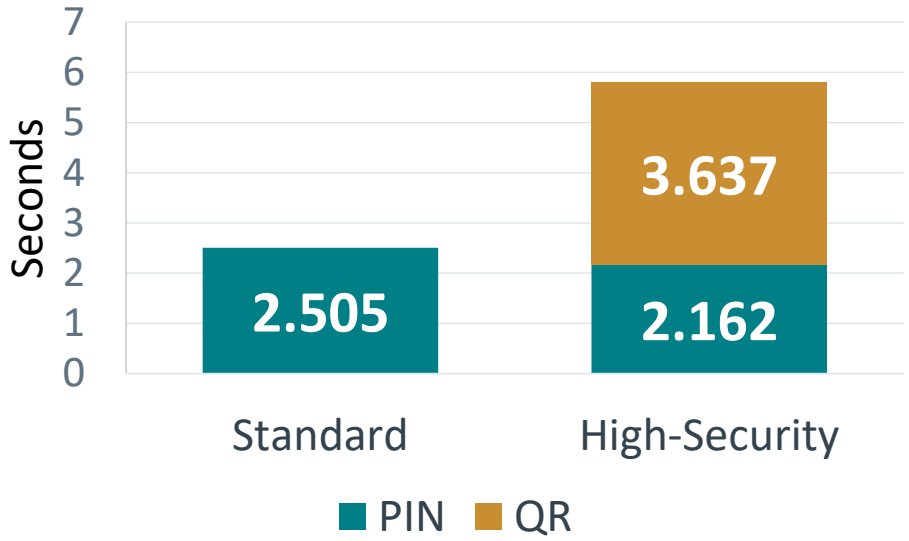
Mode	Mean	SD	DB
Fast	11.49	2.20	2.88
Standard	57.37	17.46	47.95
High-Security	54.06	6.34	44.90



70 successful entries per mode.

DB timing overlaps internal latency.

PIN/QR Interaction (s)



Metric	Standard	High
Mean	2.505	5.799
Median	2.035	4.348
SD	1.152	3.878
Maximum	6.875	22.604

Fast stage N/A. Each mode: n=70.

Manual Attendance Baseline

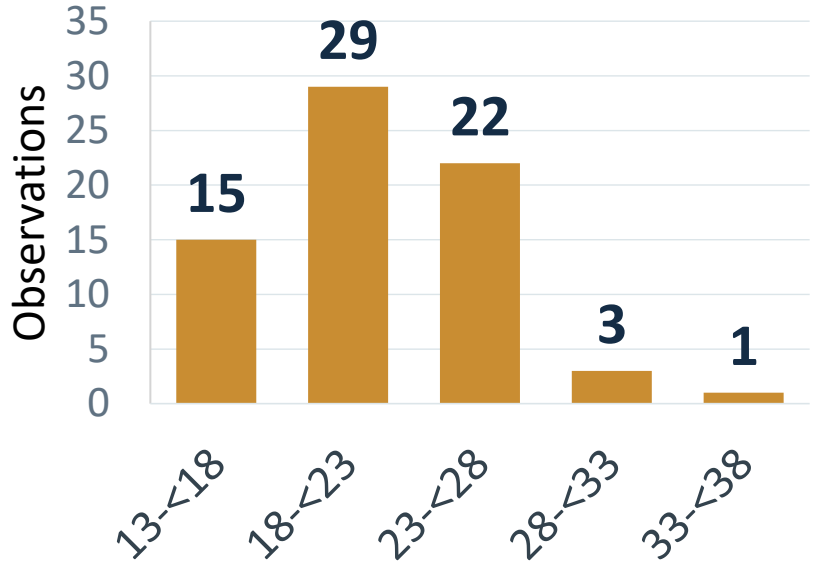
Mean: 21.803 s

SD: 4.436 s

Median: 21.135 s

Range: 13.21-37.63 s

70 active-writing observations



Task boundaries differ from system measures.

Recorded Entry Spacing and Rate

Mode	Gaps	Mean(s)	Gaps/min
Fast	66	2.730	21.98
Standard	66	8.048	7.46
High-Security	66	12.129	4.95

Four rounds: 19+19+19+9 = 66 gaps.

Complete-round throughput remains unmeasured.

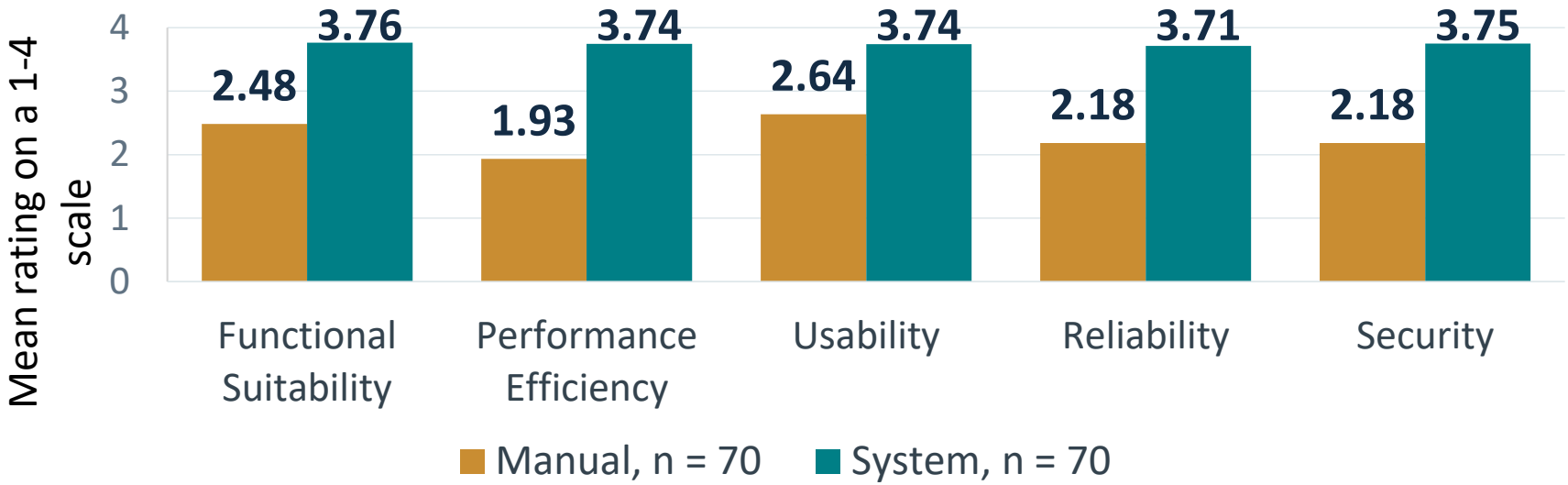
Controlled Security Results

Measure	Rate	Recorded count
Correct credential decisions	100%	222/222
Denied/all records	4.11%	18/438
Lockouts/PIN submissions	1.32%	2/152

Researcher-labelled outcomes under controlled conditions.

Demonstrations contributed to denials and lockouts.

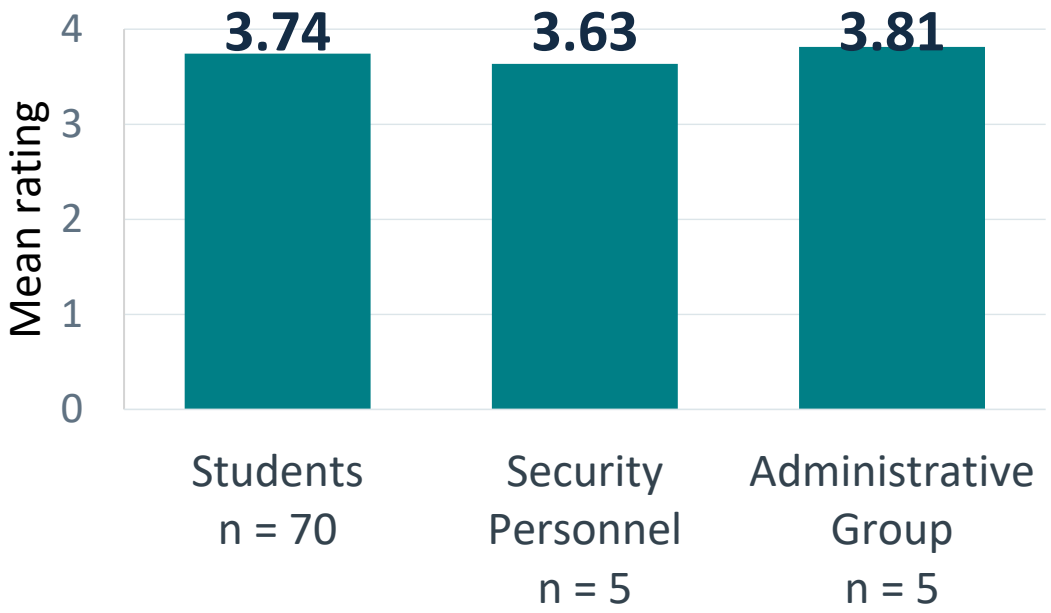
Student Survey Comparison



Performance difference: 1.811 rating points.

Different students. Categories are descriptive comparisons.

System Survey Evaluation by Group



Highly Satisfactory

All three groups

Staff minimum: 3.45

Performance Efficiency

Admin: eight categories

Other groups: five

Welch Survey Comparison

Group	n	Mean	Sample SD
Manual process	70	2.284	0.753
Proposed system	70	3.742	0.290

$t(89.076)=15.124, p<.001$

Difference=1.458, 95% CI=[1.267, 1.650]

Reject H0: Overall survey ratings differ.

Study Limitations

Physical tailgating prevention remains unverified.

Complete-task processing speed remains unmeasured.

Shared cards limit individual-identity conclusions.

Fixed mode order permits practice effects.

Different survey groups limit causal interpretation.

Small, non-random samples limit generalization.

Conclusion

Prototype integrates credential validation and attendance.

Configured modes support different verification requirements.

Controlled decisions matched researcher-labelled expectations.

Respondents reported favorable perceived system quality.

Overall student survey ratings differed significantly.

Broader effectiveness requires further evaluation.

Further Evaluation

Measure equivalent complete tasks and throughput.

Expand samples across campuses and roles.

Counterbalance the verification mode order.

Test credential sharing and outage recovery.

Evaluate later additions separately.

Complete evidence, retention, and destruction documentation.

SYSTEM DEMONSTRATION

Live Demonstration and Questions

Configured verification modes

Entry/exit timestamps and alerts

Authorized local cleanup